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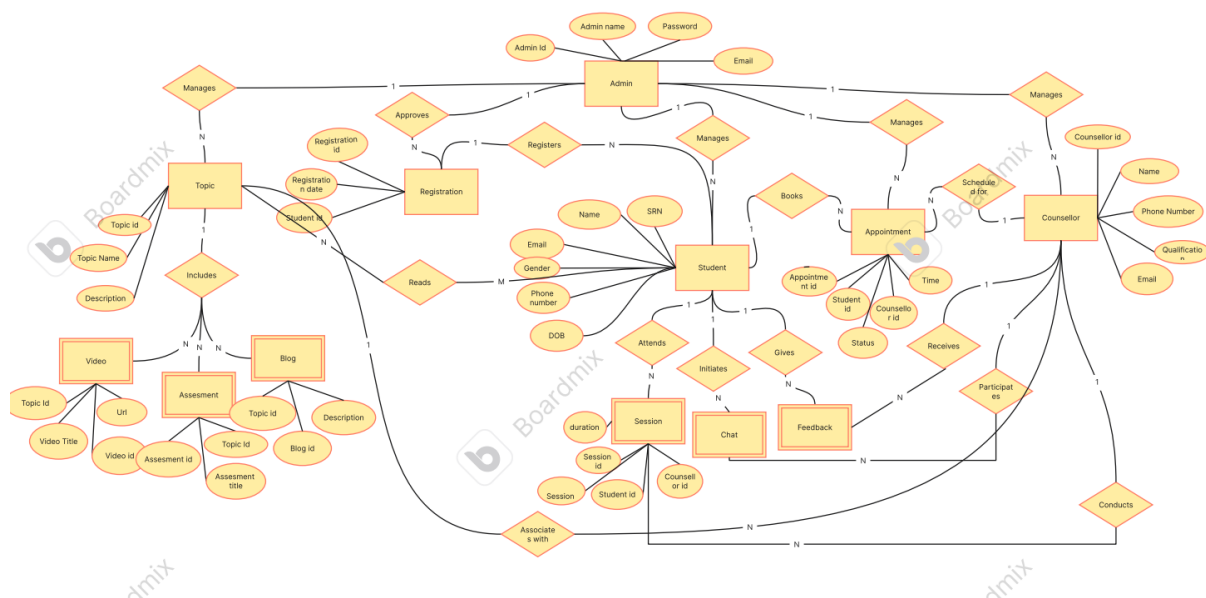
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Project Title: MindCare (Student's Mental health support website)

Software Engineering

Unit-2: Deliverables

ER Diagram:



1)Lifecycle Model : Agile model

The Agile model is suitable for this project due to its iterative nature, allowing for incremental development and frequent feedback. It enables incorporating changes based on mentor and student feedback, ensuring that the system evolves to meet real-world needs effectively.

2)Tools for each stage of Lifecycle

Stage	Tools
Planning	Jira

Design	Material UI
Version Control	GitHub
Development Tool	Visual Studio Code
Bug Tracking	Jira
Testing Tool	Postman , Selenium, Jest

3) Determine All Deliverables and Categorize Them

- **Reuse Components:**
 - **Login and Registration Module:** Can reuse authentication libraries (e.g., Firebase Auth, Passport.js).
 - **UI Framework:** Using reusable UI components from libraries like Material-UI or Bootstrap.
- **Build Components:**
 - **Customized Student and Mentor Dashboard:** Built from scratch to include unique features like topic tracking, appointments, assessments, etc.
 - **Chat Functionality:** Custom implementation to enable real-time interaction using WebSocket or Socket.IO.

Reusing authentication and UI frameworks will save development time, while custom building specific functionalities will cater to the project's unique requirements.

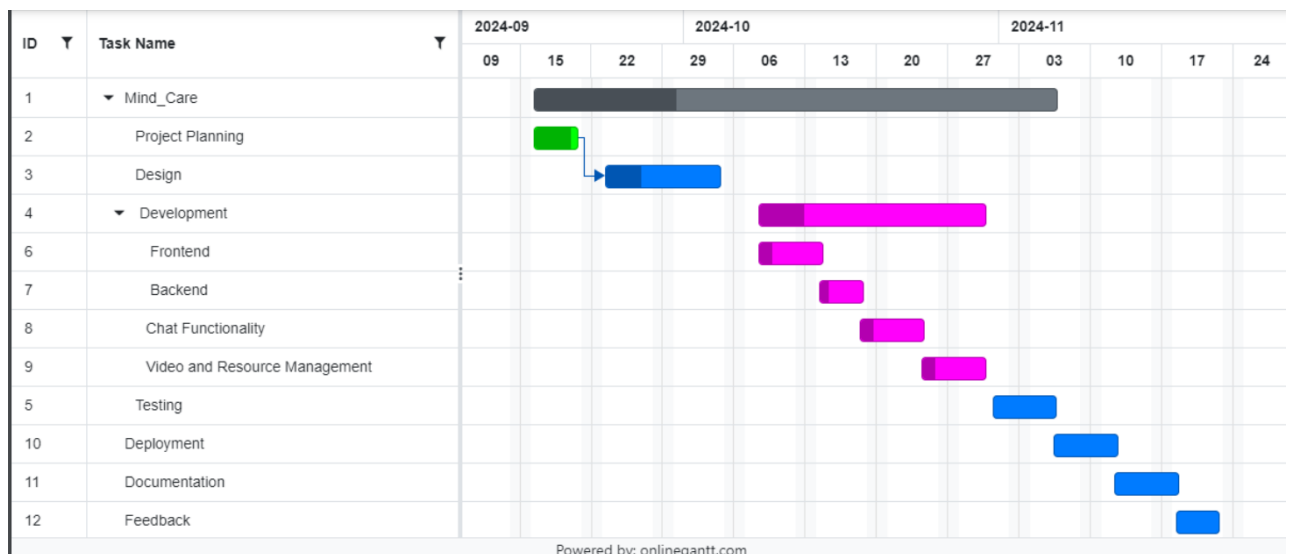
4)Work Breakdown Structure

Level	Task	Subtask
1	Planning	Requirement gathering, defining scope, scheduling
2	Design	Wireframing, UI/UX Design, Database schema design
3	Development	
	-Frontend	Setting up React environment, creating components

	-Backend	Building APIs with Express, integrating MongoDB
	-Chat Functionality	Implementing Socket.IO/WebSocket
	-Video and Resource Management	Integration with YouTube API
4	Testing	Unit testing, Integration testing, User testing
5	Deployment	Setting up deployment pipelines (CI/CD)
6	Documentation	Preparing user manuals, technical documentation
7	Feedback and iteration	Collecting feedback, making improvements

5)Effort Estimation

Gantt Chart :



Here's a breakdown of the Gantt chart for our "Mind_Care" project. Since we're a team of two, we've divided the work to make sure we finish everything within 2 months, starting on September 15th, 2024, and ending by November 3rd, 2024.

The project kicks off with Project Planning, followed by the Design phase, which only starts once the planning is done. After that, both of us will tackle

development – with Frontend and Backend happening in parallel to save time. Once the main parts of the development are done, we'll move on to Chat Functionality and Video/Resource Management.

Next, we'll dive into Testing to make sure everything works smoothly, followed by Deployment. Finally, we'll focus on Documentation and wrap things up with a Feedback phase to get input and make any last-minute improvements. By overlapping some tasks and splitting the workload, we'll be able to complete the project on time!

6) Coding Details

- **Frontend:** React.js with libraries like Material-UI for UI components.
- **Backend:** Node.js and Express for creating REST APIs.
- **Database:** MongoDB with Mongoose for schema definition and queries.
- **Authentication:** JWT (JSON Web Token) for secure authentication.
- **Chat Functionality:** Real-time chat using Socket.IO.
- **Deployment:** Use **AWS** or **Heroku** for hosting, and **Netlify** for frontend deployment